Experiment -11 Write the python program for Map Coloring to implement CSP

MAP COLORING.py - C:/Users/prana/OneDrive/Desktop/MAP COLORING.py (3.13.14)

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```
colors = ["Red", "Green", "Blue"]

graph = {
    'A': ['B', 'C'],
    'B': ['A', 'C'],
    'C': ['A', 'B']
}

assignment = {}

def safe(node, color):
    for n in graph[node]:
        if assignment.get(n) == color:
            return False
    return True

def solve(nodes):
    if not nodes:
        return True

    node = nodes[0]

    for color in colors:
        if safe(node, color):
            assignment[node] = color

            if solve(nodes[1:]):
                return True

            assignment.pop(node)

    return False

solve(['A', 'B', 'C'])

print(assignment)
```

IDLE Shell 3.13.14

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```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/prana/OneDrive/Desktop/MAP COLORING.py ==========
{'A': 'Red', 'B': 'Green', 'C': 'Blue'}
>>>
```